OPTIMAL ILLIQUIDITY

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Available at https://github.com/Tv-v-Lab/Tv-v-Reading-Group/blob/main/weekly_pre/optimal_illiquidity.pdf

HOW LIQUID SHOULD RETIREMENT SAVINGS BE?

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Retirement savings liquidity across countries

Country	System	Liquidity
US	Social Security	completely illiquid
US	401(k)/IRA	partially illiquid (10% penalty)
Australia	Super	completely illiquid
Sweden	Income Pension	completely illiquid
Sweden	ISK	completely liquid

MODEL

NAIVE HOUSEHOLD WITH UNOBSERVABLE TASTE SHOCK

Two periods: working life (period 1) and retirement (period 2)

Naive Household: chooses withdrawals ω_n from N accounts to maximize

$$\max_{(\omega_n)_{n=1}^N} \quad \theta u_1(c_1) + \beta\delta u_2(c_2), \quad s.t.$$

$$c_1 = \sum_{n=1}^{N} (1-\pi_n)\omega_n, \quad c_2 = R\sum_{n=1}^{N}(x_n - \omega_n)$$

- θ: taste shock — private info
- β: present bias — private info
- π_n: penalty rate — can be generalized
- x_n: mandatory balances

PLANNER: NO BIAS AND KNOWS TYPE DISTRIBUTIONS

Planner: chooses account balances $(x_n)_{n=1}^N$ and penalties $(\pi_n)_{n=1}^N$

$$\max_{(x_n)_{n=1}^N,(\pi_n)_{n=1}^N} \int \int \theta u_1(c_1(\theta,\beta)) + \delta u_2(c_2(\theta,\beta))\, dF(\theta) dG(\beta)$$

subject to budget balance:

$$\int \int \left(c_1(\theta,\beta) + \frac{c_2(\theta,\beta)}{R} \right) dF(\theta) dG(\beta) \leq Y$$

- F: distribution of θ
- G: distribution of β
- Y: aggregate endowment

RESULTS

SANDWICH THE GENERAL MECHANISM

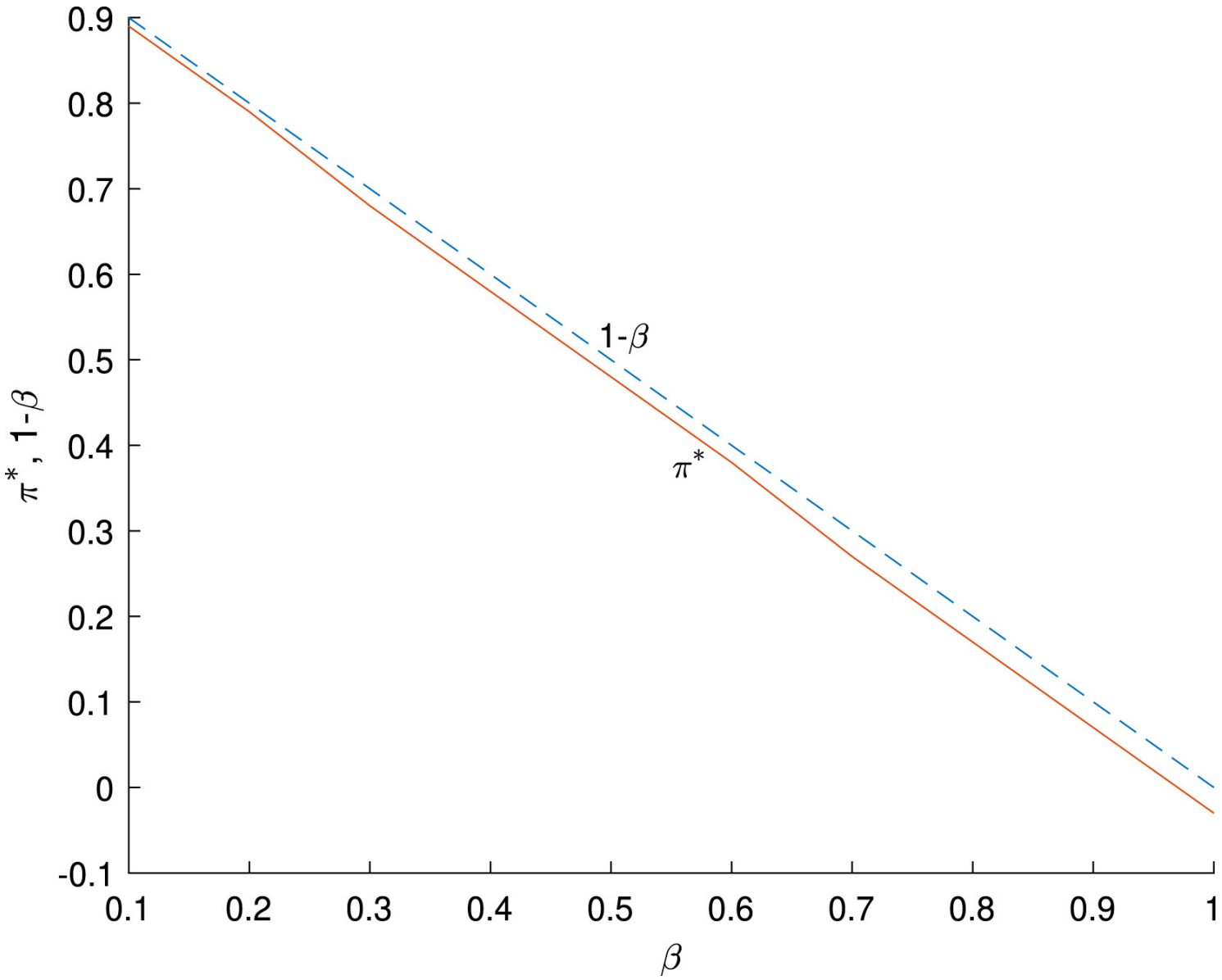
Mechanisms Comparison

Mechanism	Rules	Welfare
Linear	N accounts, linear penalties	W^L
General	Any penelty, IC constraints	W^G
Relaxed	Any penelty, less constraints	W^R

$$W^L \leq W^G \leq W^R$$

Numerically $W^R - W^L$ is small across different calibrations

HOMOGENOUS β: ONE ACCOUNT, $1 - \beta$ LINEAR PENALTY

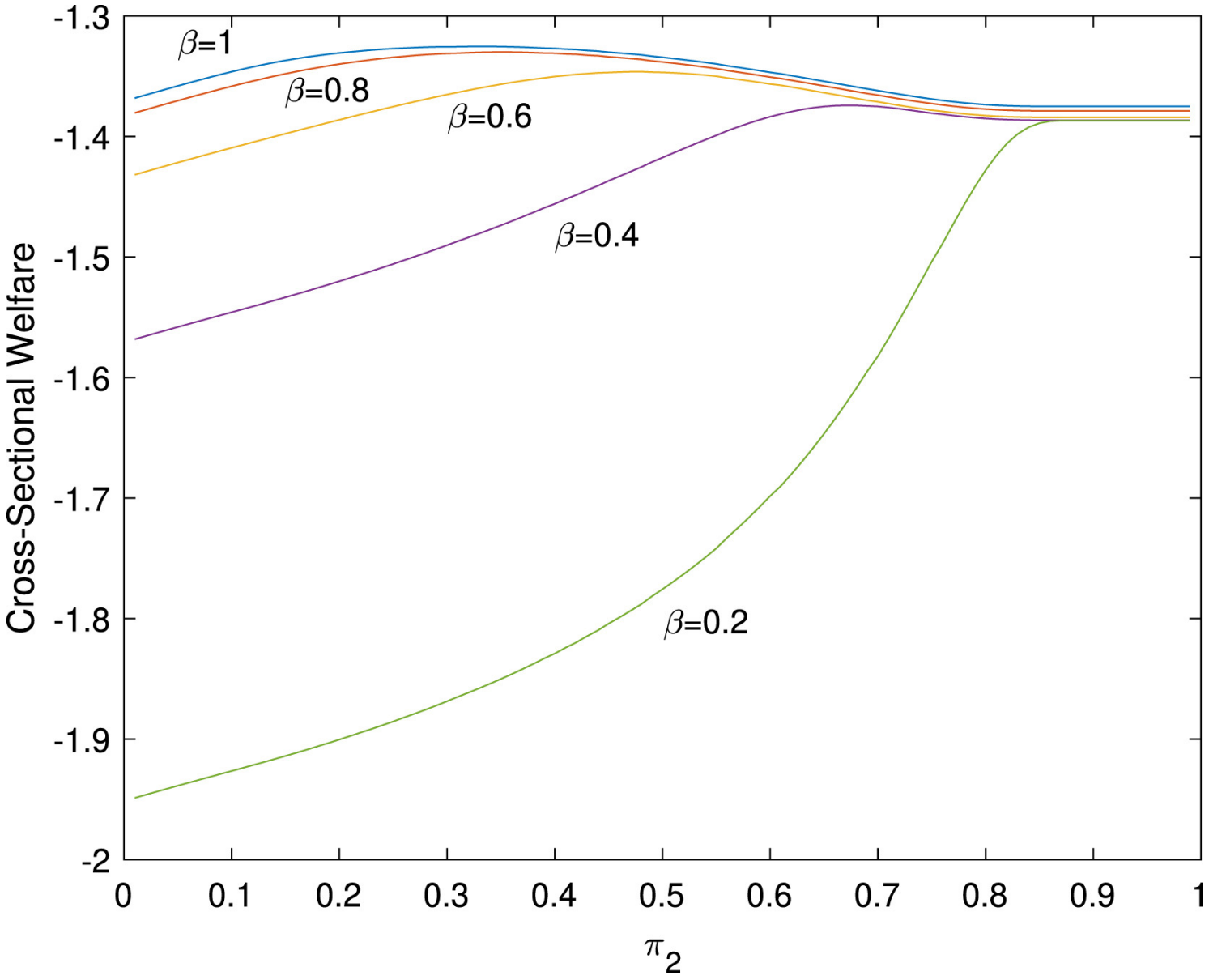


HETEROGENEOUS β: TWO ACCOUNTS, 0 AND 1 LINEAR PENALTY

Heterogeneous β: Welfare Comparison

Mechanism	Welfare (% of Y)
Linear penalty optimized, $N = 1$	3.569
Linear penalty optimized, $N = 2$	6.136
Relaxed mechanism (upper bound)	6.145
Linear penalty $\pi_1 = 0, \pi_2 = 1, N = 2$	6.105
Linear penalty $\pi_1 = 0, \pi_2 = 1, \pi_3$ optimized, $N = 3$	6.137

ILLIQUIDITY: LARGE GAINS AND SMALL LOSS



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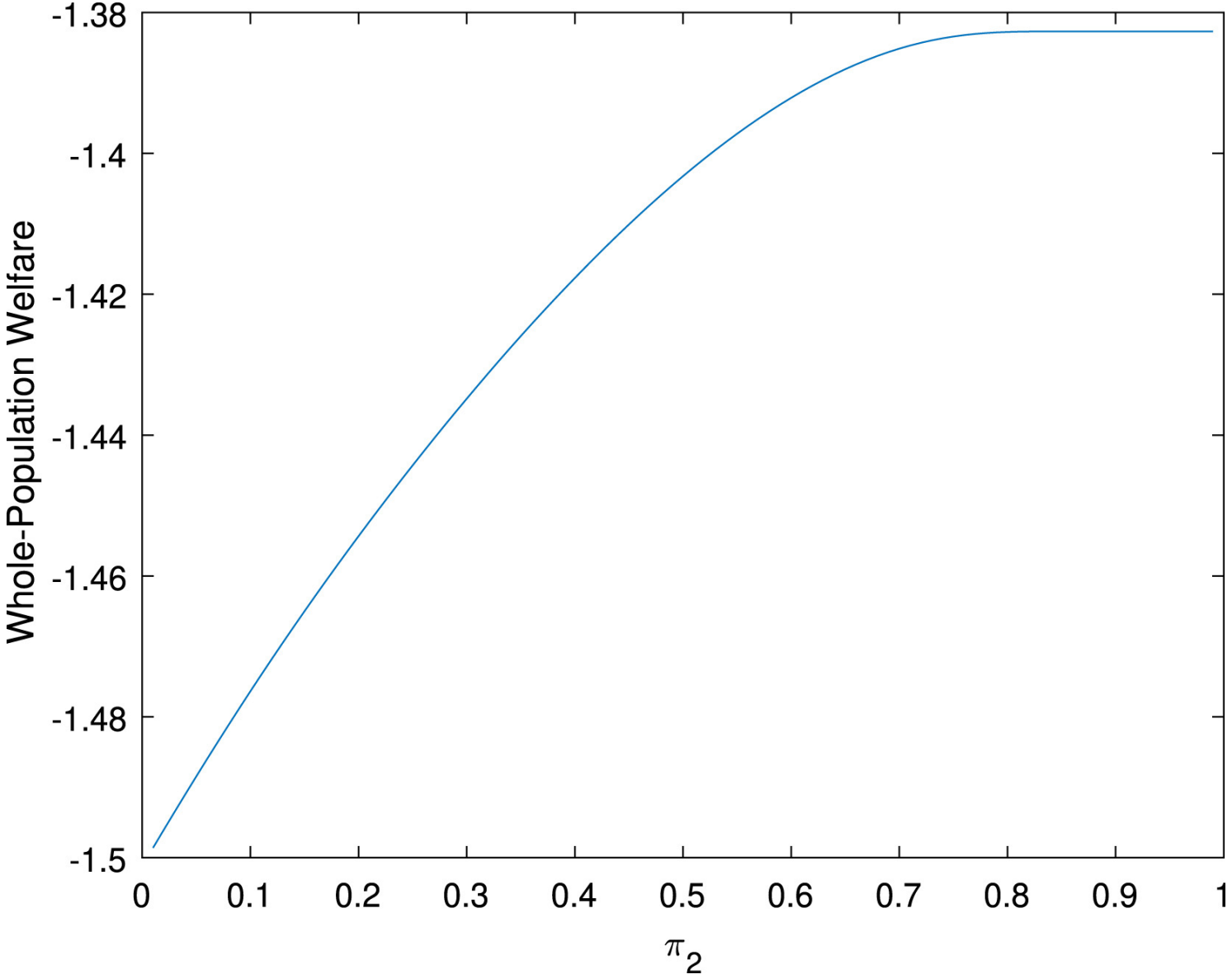

DISCUSSION AND QUESTION

LIMITATIONS

- Naivete
- Two periods only
- Main results depend on calibration
- Rationalize a status quo